

## Women's values

A scoping review of what women want from ANC and what outcomes they value informed the ANC guideline (13). Evidence showed that women from high-, medium- and low-resource settings valued having a positive pregnancy experience, the components of which included the provision of effective clinical practices (interventions and tests, including nutritional supplements), relevant and timely information (including dietary and nutritional advice) and psychosocial and emotional support, by knowledgeable, supportive and respectful health-care practitioners, to optimize maternal and newborn health (high confidence in the evidence).

## A.1: Dietary interventions

### A1.1: Counselling on healthy eating and physical activity

**RECOMMENDATION A.1.1: Counselling about healthy eating and keeping physically active during pregnancy is recommended for pregnant women to stay healthy and to prevent excessive weight gain during pregnancy. (Recommended)**

#### Remarks

- A healthy diet contains adequate energy, protein, vitamins and minerals, obtained through the consumption of a variety of foods, including green and orange vegetables, meat, fish, beans, nuts, whole grains and fruit (41).
- Stakeholders may wish to consider culturally appropriate healthy eating and exercise interventions to prevent excessive weight gain in pregnancy, particularly for populations with a high prevalence of overweight and obesity, depending on resources and women's preferences. Interventions should be woman-centred and delivered in a non-judgemental manner, and developed to ensure appropriate weight gain (see further information in points below).
- A healthy lifestyle includes aerobic physical activity and strength-conditioning exercise aimed at maintaining a good level of fitness throughout pregnancy, without trying to reach peak fitness level or train for athletic competition. Women should choose activities with minimal risk of loss of balance and fetal trauma (42).
- Most normal gestational weight gain occurs after 20 weeks of gestation and the definition of "normal" is subject to regional variations, but should take into consideration pre-pregnant body mass index (BMI). According to the Institute of Medicine classification (43), women who are underweight at the start of pregnancy (i.e. BMI < 18.5 kg/m<sup>2</sup>) should aim to gain 12.5–18 kg, women who are normal weight at the start of pregnancy (i.e. BMI 18.5–24.9 kg/m<sup>2</sup>) should aim to gain 11.5–16 kg, overweight women (i.e. BMI 25–29.9 kg/m<sup>2</sup>) should aim to gain 7–11.5 kg, and obese women (i.e. BMI > 30 kg/m<sup>2</sup>) should aim to gain 5–9 kg.
- Most evidence on healthy eating and exercise interventions comes from high-income countries (HICs), and the GDG noted that there are at least 40 ongoing trials in HICs in this field. The GDG noted that research is needed on the effects, feasibility and acceptability of healthy eating and exercise interventions in LMICs.
- Pregnancy may be an optimal time for behaviour change interventions among populations with a high prevalence of overweight and obesity, and the longer-term impact of these interventions on women, children and partners needs investigation.
- The GDG noted that a strong training package is needed for practitioners, including standardized guidance on nutrition. This guidance should be evidence-based, sustainable, reproducible, accessible and adaptable to different cultural settings.

## Summary of evidence and considerations

### Effects of diet and exercise interventions compared with no diet and exercise interventions (EB Table A.1.1)

The evidence on the effects of healthy eating and exercise interventions was derived from a Cochrane review that included 65 randomized controlled trials (RCTs), mostly conducted in HICs (44). Thirty-four trials recruited women from the general population (i.e. women of a wide range of BMIs at baseline), 24 trials recruited overweight and/or obese women and seven recruited women defined as being at high risk of gestational diabetes. In total, 49 RCTs involving 11 444 women contributed data to the review's meta-analyses. Diet interventions were defined as a special selection of food or energy intake to which a participant was restricted, which were most commonly "healthy eating" types of diets. Exercise interventions were defined by reviewers as any activity requiring physical effort, carried out to sustain or improve health or fitness, and these were either prescribed/unsupervised (e.g. 30 minutes of daily walking), supervised (e.g. a weekly supervised group exercise class) or both. These interventions were usually compared with "standard ANC" and aimed to prevent excessive gestational weight gain (EGWG).

Most trials recruited women between 10 and 20 weeks of gestation. There was substantial variation in the number of contacts (i.e. counselling/exercise sessions), type of intervention and method of delivery. Data were grouped according to the type of intervention (i.e. diet only, exercise only, diet and exercise counselling, diet and supervised exercise) and the average effects across trials were estimated using the random effects model. Separate analyses were performed according to type of intervention and the risk of weight-related complications. Most data in the overall analyses were derived from trials of combined diet and exercise interventions.

### Maternal outcomes

High-certainty evidence shows that women receiving diet and/or exercise interventions as part of ANC to prevent EGWG are less likely to experience EGWG (24 trials, 7096 women; relative risk [RR]: 0.80, 95% confidence interval [CI]: 0.73–0.87; absolute effect of 91 fewer women with EGWG per 1000 on average). Subgroup analyses were consistent with these findings.

High-certainty evidence shows that diet and/or exercise interventions have little or no effect on pre-eclampsia risk (15 trials, 5330 women; RR: 0.95, 95% CI: 0.77–1.16). However, moderate-certainty evidence indicates that diet and/or exercise interventions probably prevent hypertension in pregnancy (11 trials, 5162 women; RR: 0.70, 95% CI: 0.51–0.96).

Low-certainty evidence suggests that diet and/or exercise interventions may have little or no effect on caesarean section (28 trials, 7534 women; RR: 0.95, 95% CI: 0.88–1.03); however, low-certainty evidence from the diet and exercise counselling subgroup of trials suggests that reductions in caesarean section rates may be possible with this intervention (9 trials, 3406 women; RR: 0.87, 95% CI: 0.75–1.01). Moderate-certainty evidence indicates that diet and/or exercise interventions probably make little or no difference to induction of labour (8 trials, 3832 women; RR: 1.06, 95% CI: 0.94–1.19).

Low-certainty evidence suggests that diet and/or exercise interventions may reduce the risk of gestational diabetes mellitus (GDM) (19 trials, 7279 women; RR: 0.82, 95% CI: 0.67–1.01).

### Fetal and neonatal outcomes

Moderate-certainty evidence suggests that diet and/or exercise interventions probably prevent neonatal macrosomia (27 trials, 8598 women; RR: 0.93, 95% CI: 0.86–1.02), particularly in overweight and obese women receiving diet and exercise counselling interventions (9 trials, 3252 neonates; RR: 0.85, 95% CI: 0.73–1.00). However, moderate-certainty evidence indicates that diet and exercise interventions probably have little or no effect on neonatal hypoglycaemia (4 trials, 2601 neonates; RR: 0.95, 95% CI: 0.76–1.18) or shoulder dystocia (4 trials, 3253 neonates; RR: 1.02, 95% CI: 0.57–1.83). Low-certainty evidence suggests that neonatal respiratory morbidity may occur less frequently with diet and exercise counselling interventions than controls, particularly among overweight and obese women (2 studies, 2256 women; RR: 0.47, 95% CI: 0.26–0.85).

Low-certainty evidence suggests that diet and/or exercise interventions may have little or no effect on preterm birth (16 trials, 5923 women; RR: 0.91, 95% CI: 0.68–1.22), and the evidence on low-birth-weight neonates is very uncertain. Perinatal mortality was not reported in the review.

### Additional considerations

- High-certainty evidence from the review also shows that low gestational weight gain is more likely to occur with these interventions (11 trials, 4422 women; RR: 1.14, CI: 1.02–1.27); the clinical relevance of this finding is not known.
- The effects, acceptability and feasibility of diet and exercise interventions in LMICs has not been established.

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

Cost implications of diet and exercise interventions for health services are highly variable. For example, supervised diet and exercise interventions can have high associated costs, mainly due to staff costs for time spent supervising, while counselling interventions might have relatively low costs. For pregnant women, the interventions might also have resource implications in terms of transport costs, time off work and child-minding costs, particularly if the intervention requires additional antenatal visits.

### Equity

Most of the evidence came from trials conducted in HICs. Recent studies have reported a shift in the burden of overweight and obesity from advantaged to disadvantaged populations (32). Such a trend increases the risk of associated pregnancy complications, as well as cardiometabolic problems, among pregnant women from disadvantaged

populations. These risks might be further exacerbated among women in low-resource community settings, as these settings may not be equipped to deal with complications.

### Acceptability

Qualitative evidence indicates that women in a variety of settings tend to view ANC as a source of knowledge and information and that they generally appreciate any advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22). It also suggests that women may be less likely to engage with health services if advice is delivered in a hurried or didactic manner (high confidence in the evidence) (22). Therefore, these types of interventions are more likely to be acceptable if the interventions are delivered in an unhurried and supportive way, which may also facilitate better engagement with ANC services. Qualitative evidence on health-care providers’ views of ANC suggests that they may be keen to offer general health-care advice and specific pregnancy-related information (low confidence in the evidence) but they sometimes feel they do not have the appropriate training and lack the resources and time to deliver the service in the informative, supportive and caring manner that women want (high confidence in the evidence) (45).

### Feasibility

In a number of LMIC settings, providers feel that a lack of resources may limit implementation of recommended interventions (high confidence in the evidence) (45).

## A.1.2: Nutrition education on energy and protein intake

**RECOMMENDATION A.1.2: In undernourished populations, nutrition education on increasing daily energy and protein intake is recommended for pregnant women to reduce the risk of low-birth-weight neonates.** (*Context-specific recommendation*)

### Remarks

- Undernourishment is usually defined by a low BMI (i.e. being underweight). For adults, a 20–39% prevalence of underweight women is considered a high prevalence of underweight and 40% or higher is considered a very high prevalence (46). Mid-upper arm circumference (MUAC) may also be useful to identify protein–energy malnutrition in individual pregnant women and to determine its prevalence in this population (31). However, the optimal cut-off points may need to be determined for individual countries based on context-specific cost–benefit analyses (31).
- Anthropometric characteristics of the general population are changing, and this needs to be taken into account by regularly reassessing the prevalence of undernutrition to ensure that the intervention remains relevant.
- The GDG noted that a strong training package is needed for practitioners, including standardized guidance on nutrition. This guidance should be evidence-based, sustainable, reproducible, accessible and adaptable to different cultural settings.
- Stakeholders might wish to consider alternative delivery platforms (e.g. peer counsellors, media reminders) and task shifting for delivery of this intervention.
- Areas that are highly food insecure or those with little access to a variety of foods may wish to consider additional complementary interventions, such as distribution of balanced protein and energy supplements (see Recommendation A.1.3).

## Summary of evidence and considerations

### Effects of nutritional education to increase energy and protein intake versus no nutritional education intervention (EB Table A.1.2)

Evidence on the effects of nutritional education was derived from a Cochrane review (47). Five trials conducted between 1975 and 2013 in Bangladesh, Greece and the USA, involving 1090 pregnant women, contributed data to this comparison. Nutritional education interventions were delivered one-to-one or in group classes and included education to improve the “quality” of diet, increase energy and protein intake, or improve knowledge of the nutritional value of different foods, including energy, protein, vitamins and iron. The Bangladesh study also involved cookery demonstrations.

### Maternal outcomes

Evidence on gestational weight gain was of very low certainty. There was no other evidence available on maternal outcomes in the review for this comparison.

### Fetal and neonatal outcomes

Low-certainty evidence shows that antenatal dietary education may reduce low-birth-weight neonates (300 women; RR: 0.04, 95% CI: 0.01–0.14), but may

have little or no effect on small-for-gestational-age (SGA) neonates (2 trials, 449 women; RR: 0.46, 95% CI: 0.21–0.98), stillbirths (1 trial, 431 women; RR: 0.37, 95% CI: 0.07–1.90) or neonatal deaths (1 trial, 448 women; RR: 1.28, 95% CI: 0.35–4.72). Evidence on preterm birth was judged to be of very low certainty.

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

Resource costs are variable and mainly related to staffing and counselling time.

### Equity

In many LMICs, pregnancy outcomes and ANC coverage are worse among women who are poor, least educated and residing in rural areas (29). Many low-income countries still struggle with widespread poverty and hunger, particularly among rural populations (48). Findings from a study of antenatal food supplementation and micronutrient supplements in rural Bangladesh suggest that food supplementation interventions might be associated with better ANC adherence among women with

less education but not among those with more education (49). Therefore, providing antenatal food supplements could help to address inequalities by improving maternal nutritional status and increasing ANC coverage among disadvantaged women.

### Acceptability

Qualitative evidence indicates that women in a variety of settings tend to view ANC as a source of knowledge and information and that they generally appreciate any advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22). It also suggests that women may be less likely to engage with health services if advice is delivered in a hurried or didactic manner (high confidence in the evidence) (22). Therefore, these types of interventions are more likely to be acceptable if the interventions are delivered in an unhurried

and supportive way, which may also facilitate better engagement with ANC services. Qualitative evidence on health-care providers' views of ANC suggests that they may be keen to offer general health-care advice and specific pregnancy-related information (low confidence in the evidence) but they sometimes feel they do not have the appropriate training and lack the resources and time to deliver the service in the informative, supportive and caring manner that women want (high confidence in the evidence) (45).

### Feasibility

In a number of LMIC settings, providers feel that a lack of resources may limit implementation of recommended interventions (high confidence in the evidence) (45).

### A.1.3: Energy and protein dietary supplements

**RECOMMENDATION A.1.3: In undernourished populations, balanced energy and protein dietary supplementation is recommended for pregnant women to reduce the risk of stillbirths and small-for-gestational-age neonates.** (*Context-specific recommendation*)

#### Remarks

- The GDG stressed that this recommendation is for populations or settings with a high prevalence of undernourished pregnant women, and not for individual pregnant women identified as being undernourished.
- Undernourishment is usually defined by a low BMI (i.e. being underweight). For adults, a 20–39% prevalence of underweight women is considered a high prevalence of underweight and 40% or higher is considered a very high prevalence (46). MUAC may also be useful to identify protein-energy malnutrition in individual pregnant women and to determine its prevalence in this population (31). However, the optimal cut-off points may need to be determined for individual countries based on context-specific cost-benefit analyses (31).
- Establishment of a quality assurance process is important to guarantee that balanced energy and protein food supplements are manufactured, packaged and stored in a controlled and uncontaminated environment. The cost and logistical implications associated with balanced energy and protein supplements might be mitigated by local production of supplements, provided that a quality assurance process is established.
- A continual, adequate supply of supplements is required for programme success. This requires a clear understanding and investment in procurement and supply chain management.
- Programmes should be designed and continually improved based on locally generated data and experiences. Examples relevant to this guideline include:
  - Improving delivery, acceptability and utilization of this intervention by pregnant women (i.e. overcoming supply and utilization barriers).
  - Distribution of balanced energy and protein supplements may not be feasible only through the local schedule of ANC visits; additional visits may need to be scheduled. The costs related to these additional visits should be considered. In the absence of antenatal visits, too few visits, or when the first visit comes too late, consideration should be given to alternative platforms for delivery (e.g. community health workers, task shifting in specific settings).
  - Values and preferences related to the types and amounts of balanced energy and protein supplements may vary.
- Monitoring and evaluation should include evaluation of household-level storage facilities, spoilage, wastage, retailing, sharing and other issues related to food distribution.
- Each country will need to understand the context-specific etiology of undernutrition at the national and sub-national levels. For instance, where seasonality is a predictor of food availability, the programme should consider this and adapt to the conditions as needed (e.g. provision of more or less food of different types in different seasons). In addition, a better understanding is needed of whether alternatives to energy and protein supplements – such as cash or vouchers, or improved local and national food production and distribution – can lead to better or equivalent results.
- Anthropometric characteristics of the general population are changing, and this needs to be taken into account to ensure that only those women who are likely to benefit (i.e. only undernourished women) are included.
- The GDG noted that it is not known whether there are risks associated with providing this intervention to women with a high BMI.

## Summary of evidence and considerations

### Effects of balanced energy and protein supplements compared with no supplements or placebo (EB Table A.1.3)

Evidence on the effects of balanced energy and protein supplements compared with no supplementation or placebo was derived from a Cochrane review (47). Twelve trials, involving 6705 women, were included in this comparison. Most data were derived from trials conducted in LMICs, including Burkina Faso, Colombia, Gambia, Ghana, India, Indonesia, South Africa and Taiwan, China. The balanced energy and protein supplements used were in various forms, including fortified beverages, biscuits and powders.

### Maternal outcomes

The only maternal outcome reported for this comparison in the review, of those outcomes prioritized for this guideline, was pre-eclampsia. However, the evidence on this outcome, based on two small trials, was assessed as very uncertain.

### Fetal and neonatal outcomes

Moderate-certainty evidence shows that balanced energy and protein supplementation probably reduces SGA neonates (7 trials, 4408 women; RR: 0.79, 95% CI: 0.69–0.90) and stillbirths (5 trials, 3408 women; RR: 0.60, 95% CI: 0.39–0.94), but probably has no effect on preterm birth (5 trials, 3384 women; RR: 0.96, 95% CI: 0.80–1.16). Low-certainty evidence suggests that it may have little or no effect on neonatal deaths (5 trials, 3381 women; RR: 0.68, 95% CI: 0.43–1.07). Low birth weight was not reported for this comparison in the review.

### Additional considerations

- In the review, mean birth weight (in grams) was reported and the findings favoured the balanced energy and protein supplementation group (11 trials, 5385 neonates; mean difference [MD]: 40.96, 95% CI: 4.66–77.26). This evidence was graded as moderate-quality evidence in the review (47).

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

The cost of balanced energy and protein supplements is relatively high. There may also be cost implications with respect to transport, storage and training.

### Equity

In many LMICs, pregnancy outcomes and ANC coverage are worse among women who are poor, least educated and residing in rural areas (29). Many low-income countries still struggle with widespread poverty and hunger, particularly among rural populations (48). Findings from a study of antenatal food supplementation and micronutrient supplements in rural Bangladesh suggest that food supplementation interventions might be associated with better ANC adherence among women with less education but not among those with more education (49). Therefore, providing antenatal food supplements could help to address inequalities by improving maternal nutritional status and increasing ANC coverage among disadvantaged women.

### Acceptability

Qualitative evidence indicates that women in a variety of settings tend to view ANC as a source of knowledge and information and that they generally appreciate any advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22). It also suggests that women may be less likely to engage with health services if advice is delivered in a hurried or didactic manner (high confidence in the evidence) (22). Therefore, these types of interventions are more likely to be acceptable if the interventions are delivered in an unhurried and supportive way, which may also facilitate better engagement with ANC services. Qualitative evidence on health-care providers’ views of ANC suggests that they may be keen to offer general health-care advice and specific pregnancy-related information (low confidence in the evidence) but they sometimes feel they do not have the appropriate training and lack the resources and time to deliver the service in the informative, supportive and caring manner that women want (high confidence in the evidence) (45).

### Feasibility

Providing balanced protein and energy supplements may be associated with logistical issues, as supplements are bulky and will require adequate transport and storage facilities to ensure continual supplies. Qualitative evidence from LMIC settings indicates that providers feel that a lack of resources may limit implementation of recommended interventions (high confidence in the evidence) (45).



## A.1.4: High-protein supplements

**RECOMMENDATION A.1.4: In undernourished populations, high-protein supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes. (Not recommended)**

### Remarks

- The GDG noted that there is insufficient evidence on the benefits, if any, of high-protein supplementation.
- Further research on the effects of high-protein supplements in undernourished populations is not considered a research priority.

## Summary of evidence and considerations

### Effects of high-protein supplementation compared with controls (EB Table A.1.4)

Evidence on the effects of high-protein supplementation was derived from the same Cochrane review as for Recommendations A.1.2 and A.1.3 (47). The review included one trial of high-protein supplementation compared with a micronutrient supplement conducted in the 1970s, involving 1051 low-income, black women in the USA.

### Maternal outcomes

None of the outcomes prioritized for this guideline were reported for this comparison in the review.

### Fetal and neonatal outcomes

High-certainty evidence shows that high-protein supplementation increases SGA neonates (1 trial, 505 neonates; RR: 1.58, 95% CI: 1.03–2.41). Moderate-certainty evidence indicates that high-protein supplementation probably has little or no effect on preterm birth (1 study, 505 women; RR: 1.14, 95% CI: 0.83–1.56). Low-certainty evidence suggests that high-protein supplementation may have little or no effect on stillbirths (1 trial, 529 babies; RR: 0.81, 95% CI: 0.31–2.15; certainty of evidence downgraded due to imprecision) and neonatal deaths (1 trial, 529 neonates; RR: 2.78, 95% CI: 0.75–10.36).

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

The cost of high-protein supplements is relatively high. There may also be cost implications with respect to transport, storage and training.

### Equity

In many LMICs, pregnancy outcomes and ANC coverage are worse among women who are poor, least educated and residing in rural areas (29). Many low-income countries still struggle with widespread poverty and hunger, particularly among rural populations (48). Therefore, providing antenatal food supplements could help to address inequalities by improving maternal nutritional status and increasing ANC coverage among disadvantaged women.

### Acceptability

Qualitative evidence indicates that women in a variety of settings tend to view ANC as a source of knowledge and information and that they generally appreciate any advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22). It also suggests that women may be less likely to engage with health services if advice is delivered in a hurried or didactic manner (high confidence in the evidence) (22). Qualitative evidence on health-care providers’ views of ANC suggests that they may be keen to offer general health-care advice and specific pregnancy-related information (low confidence in the evidence) but they sometimes feel they do not have the appropriate training and lack the resources and time to deliver the service in the informative, supportive and caring manner that women want (high confidence in the evidence) (45).

### Feasibility

Providing high-protein supplements may be associated with logistical issues, as supplements are bulky and will require adequate transport and storage facilities to ensure continual supplies. Qualitative evidence from LMIC settings indicates that providers feel that a lack of resources may limit implementation of recommended interventions (high confidence in the evidence) (45).



## A.2: Iron and folic acid supplements

### A.2.1: Daily iron and folic acid supplements

**RECOMMENDATION A.2.1: Daily oral iron and folic acid supplementation with 30 mg to 60 mg of elemental iron<sup>a</sup> and 400 µg (0.4 mg) folic acid<sup>b</sup> is recommended for pregnant women to prevent maternal anaemia, puerperal sepsis, low birth weight, and preterm birth.<sup>c</sup>**  
(Recommended)

#### Remarks

- This recommendation supersedes the 2012 WHO *Guideline: daily iron and folic acid supplementation in pregnant women* (36) and should be considered alongside Recommendation A.2.2 on intermittent iron.
- In settings where anaemia in pregnant women is a severe public health problem (i.e. where at least 40% of pregnant women have a blood haemoglobin [Hb] concentration < 110 g/L), a daily dose of 60 mg of elemental iron is preferred over a lower dose.
- In the first and third trimesters, the Hb threshold for diagnosing anaemia is 110 g/L; in the second trimester, the threshold is 105 g/L (50).
- If a woman is diagnosed with anaemia during pregnancy, her daily elemental iron should be increased to 120 mg until her Hb concentration rises to normal (Hb 110 g/L or higher) (34, 51). Thereafter, she can resume the standard daily antenatal iron dose to prevent recurrence of anaemia.
- Effective communication with pregnant women about diet and healthy eating – including providing information about food sources of vitamins and minerals, and dietary diversity – is an integral part of preventing anaemia and providing quality ANC.
- Effective communication strategies are vital for improving the acceptability of, and adherence to, supplementation schemes.
- Stakeholders may need to consider ways of reminding pregnant women to take their supplements and of assisting them to manage associated side-effects.
- In areas with endemic infections that may cause anaemia through blood loss, increased red cell destruction or decreased red cell production, such as malaria and hookworm, measures to prevent, diagnose and treat these infections should be implemented.
- Oral supplements are available as capsules or tablets (including soluble tablets, and dissolvable and modified-release tablets) (52). Establishment of a quality assurance process is important to guarantee that supplements are manufactured, packaged and stored in a controlled and uncontaminated environment (53).
- A better understanding of the etiology of anaemia (e.g. malaria endemicity, haemoglobinopathies) and the prevalence of risk factors is needed at the country level, to inform context-specific adaptations of this recommendation.
- Standardized definitions of side-effects are needed to facilitate monitoring and evaluation.
- Development and improvement of integrated surveillance systems are needed to link the assessment of anaemia and iron status at the country level to national and global surveillance systems.
- To reach the most vulnerable populations and ensure a timely and continuous supply of supplements, stakeholders may wish to consider task shifting the provision of iron supplementation in community settings with poor access to health-care professionals (see Recommendation E.6.1, in section E: Health systems interventions to improve the utilization and quality of ANC).

a The equivalent of 60 mg of elemental iron is 300 mg of ferrous sulfate heptahydrate, 180 mg of ferrous fumarate or 500 mg of ferrous gluconate.

b Folic acid should be commenced as early as possible (ideally before conception) to prevent neural tube defects.

c This recommendation supersedes the previous WHO recommendation found in the 2012 *Guideline: daily iron and folic acid supplementation in pregnant women* (36).

## Summary of evidence and considerations

### Effects of any daily iron and folic acid supplements compared with no daily iron and folic acid supplements (EB Table A.2.1)

The evidence on the effects of daily iron and/or folic acid was derived from a Cochrane review of 61 trials conducted in low-, middle- and high-income countries (54). Twenty-three trials were conducted in countries with some malaria risk, of which two reported malaria outcomes. Overall, 44 trials involving 43 274 women contributed data to the review's meta-analyses. The trials compared daily oral iron supplementation, with or without folic acid or other vitamin and mineral supplements, with various control groups (folic acid only, placebo, no intervention, other vitamin and mineral supplements without iron or folic acid). Most of the evidence was derived from studies comparing iron supplementation with no iron supplementation. In most trials, women began taking supplements before 20 weeks of gestation and continued taking supplements until delivery. The most commonly used dose of elemental iron was 60 mg daily (range: 30–240 mg) and that of folic acid was 400 µg daily.

### Maternal outcomes

Anaemia was reported in many different ways and at different time points during pregnancy and the puerperium. Low-certainty evidence shows that daily iron supplementation may reduce the risk of anaemia at term (defined as blood Hb concentration < 110 g/L at 37 weeks of gestation or later) (14 trials, 2199 women; RR: 0.30, 95% CI: 0.19–0.46) and severe postpartum anaemia (defined as Hb < 80 g/L) (8 trials, 1339 women; RR: 0.04, 95% CI: 0.01–0.28).

Low-certainty evidence also shows that daily iron supplementation may increase maternal Hb concentrations at or near term (34 weeks of gestation or more) (19 trials, 3704 women; MD: 8.88 g/L higher, 95% CI: 6.96–10.8 g/L) and may increase the proportion of women with a high maternal Hb at or near term (Hb > 130 g/L at 34 weeks of gestation or later) (8 trials, 2156 women; RR: 3.07, 95% CI: 1.18–8.02).

Regarding maternal morbidity, moderate-certainty evidence shows that daily iron supplementation probably reduces the risk of maternal puerperal infections (4 trials, 4374 women; RR: 0.68, 95% CI: 0.5–0.92). Low-certainty evidence shows that daily

iron supplementation may have little or no effect on pre-eclampsia (4 trials, 1704 women; RR: 1.63, 95% CI: 0.87–3.07) and antepartum haemorrhage (2 trials, 1157 women; RR: 1.48, 95% CI: 0.51–4.31), and moderate-certainty evidence shows that it probably has little or no effect on postpartum haemorrhage (4 trials, 1488 women; RR: 0.93, 95% CI: 0.59–1.49). Evidence on other morbidity outcomes, including placental abruption and blood transfusions, is of very low certainty.

Low-certainty evidence shows that daily iron supplementation may have little or no effect on maternal mortality (2 trials, 12 560 women; RR: 0.33, 95% CI: 0.01–8.19). Women's satisfaction was evaluated in one small trial (49 women), which found little difference between daily iron and control groups.

**Side-effects:** Moderate-certainty evidence indicates that daily iron supplementation probably has little or no effect on the risk of experiencing any side-effect (11 trials, 2425 women; RR: 1.29, 95% CI: 0.83–2.02), and that it may have little or no effect on constipation (4 trials, 1495 women; RR: 0.95, 95% CI: 0.62–1.43), heartburn (3 trials, 1323 women; RR: 1.19, 95% CI: 0.86–1.66) and vomiting (4 trials, 1392 women; RR: 0.88, 95% CI: 0.59–1.30). Evidence that daily iron has little or no effect on nausea is of low certainty (4 trials, 1377 women; RR: 1.21, 95% CI: 0.72–2.03). High-certainty evidence shows that diarrhoea is less common with daily iron supplements (3 trials, 1088 women; RR: 0.55, 95% CI: 0.32–0.93).

### Fetal and neonatal outcomes

Low-certainty evidence shows that daily iron may reduce the risk of low-birth-weight neonates (< 2500 g) (11 trials, 17 613 neonates; RR: 0.84, 95% CI: 0.69–1.03). High-certainty evidence shows that it does not reduce the risk of preterm birth before 37 weeks of gestation (13 trials, 19 286 women; RR: 0.93, 95% CI: 0.84–1.03), but it does reduce the risk of very preterm birth (i.e. less than 34 weeks of gestation) (5 trials, 3749 women; RR: 0.51, 95% CI: 0.29–0.91).

Low-certainty evidence suggests that daily iron may have little or no effect on congenital anomalies (4 trials, 14 636 neonates; RR: 0.88, 95% CI: 0.58–1.33). Moderate-certainty evidence indicates that daily iron probably has little or no effect on neonatal deaths (4 trials, 16 603 neonates; RR: 0.91, 95% CI: 0.71–1.18). Neonatal infections and SGA were not reviewed as outcomes.

### Additional considerations

- Evidence from subgroups tended to be consistent with the overall findings for the main outcomes. More details can be found in the Web supplement (EB Table A.2.1).

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

Daily iron and folic acid supplements are relatively low cost, at less than 1 United States dollar (US\$ 1) per pregnant woman (27).

### Equity

Iron deficiency and parasitic infections are more common in LMICs and disadvantaged populations. Poor, rural and least-educated populations also experience the highest maternal, infant and child mortality (29). Increasing coverage of effective nutritional interventions to prevent anaemia, particularly among disadvantaged populations,

might help to address maternal and newborn health inequalities.

### Acceptability

Qualitative evidence suggests that the availability of iron supplements may actively encourage women to engage with ANC providers (low confidence in the evidence) (22). However, where there are additional costs associated with supplementation or where the supplements may be unavailable (because of resource constraints) women are less likely to engage with ANC services (high confidence in the evidence). Lower doses of iron may be associated with fewer side-effects and therefore may be more acceptable to women than higher doses.

### Feasibility

Qualitative evidence about the views of health-care providers suggests that resource constraints, both in terms of the availability of the supplements and the lack of suitably trained staff to deliver them, may limit implementation (high confidence in the evidence) (45).

## A.2.2: Intermittent iron and folic acid supplements

**RECOMMENDATION A.2.2: Intermittent oral iron and folic acid supplementation with 120 mg of elemental iron<sup>a</sup> and 2800 µg (2.8 mg) of folic acid once weekly is recommended for pregnant women to improve maternal and neonatal outcomes if daily iron is not acceptable due to side-effects, and in populations with an anaemia prevalence among pregnant women of less than 20%. (Context-specific recommendation)**

### Remarks

- This recommendation supersedes the previous WHO recommendation in the 2012 Guideline: intermittent iron and folic acid supplementation in non-anaemic pregnant women (55) and should be considered alongside Recommendation A.1.1.
- In general, anaemia prevalence of less than 20% is classified as a mild public health problem (33).
- Before commencing intermittent iron supplementation, accurate measurement of maternal blood Hb concentrations is needed to confirm the absence of anaemia. Therefore, this recommendation may require a strong health system to facilitate accurate Hb measurement and to monitor anaemia status throughout pregnancy.
- If a woman is diagnosed with anaemia (Hb < 110 g/L) during ANC, she should be given 120 mg of elemental iron and 400 µg (0.4 mg) of folic acid daily until her Hb concentration rises to normal (Hb 110 g/L or higher) (34, 51). Thereafter, she can continue with the standard daily antenatal iron and folic acid dose (or the intermittent regimen if daily iron is not acceptable due to side-effects) to prevent recurrence of anaemia.
- Stakeholders may need to consider ways of reminding pregnant women to take their supplements on an intermittent basis and of assisting them to manage associated side-effects.

a The equivalent of 120 mg of elemental iron is 600 mg of ferrous sulfate heptahydrate, 360 mg of ferrous fumarate or 1000 mg of ferrous gluconate.

## Summary of evidence and considerations

### Effects of intermittent iron and folic acid supplements compared with daily iron and folic acid supplements (EB Table A.2.2)

The evidence on the effects of intermittent iron and folic acid was derived from a Cochrane review that included 27 trials from 15 countries; however, only 21 trials (involving 5490 women) contributed data to the review's meta-analyses (56). All trials were conducted in LMICs with some degree of malaria risk (Argentina, Bangladesh, China, Guatemala, India, Indonesia, the Islamic Republic of Iran, Malawi, Malaysia, Mexico, Pakistan, Republic of Korea, Sri Lanka, Thailand and Viet Nam); however, only one trial specifically reported that it was conducted in a malaria-endemic area.

Most of the intermittent iron regimens involved women taking weekly supplements, most commonly 120 mg elemental iron per week (range: 80–200 mg weekly), which was compared with daily regimens, most commonly 60 mg elemental iron daily (range: 40–120 mg daily). Where folic acid was also provided in the trials, it was administered weekly in the intermittent supplement groups (range: 400–3500 µg weekly) compared with the usual standard daily dose for control groups.

### Maternal outcomes

Anaemia was reported in different ways across trials. Low-certainty evidence suggests there may be little or no difference between intermittent and daily iron supplementation in the effect on anaemia at term (4 trials, 676 women; RR: 1.22, 95% CI: 0.84–1.80). Moderate-certainty evidence shows that anaemia at or near term (defined as a Hb of < 110 g/L at 34 weeks of gestation or later) probably occurs more frequently with intermittent than daily iron supplementation (8 trials, 1385 women; RR: 1.66, 95% CI: 1.09–2.53), and that intermittent iron supplementation is probably less likely to be associated with a Hb concentration of more than 130 g/L than daily iron (15 trials, 2616 women; RR: 0.53, 95% CI: 0.38–0.74). No events of severe anaemia occurred in either group in six trials reporting this outcome (1240 women). The evidence on mean Hb concentrations at or near term and severe postpartum anaemia is of very low certainty.

Limited evidence on maternal morbidity from one small trial (110 women) was assessed as

very uncertain. Maternal infections and maternal satisfaction were not evaluated in the review.

**Side-effects:** Moderate-certainty evidence shows that intermittent iron supplementation is probably less commonly associated with nausea than daily iron supplementation (7 trials, 1034 women; RR: 0.60, 95% CI: 0.37–0.97). However, the evidence on other specific side-effects (constipation, diarrhoea, heartburn or vomiting) or any side-effect is of very low certainty.

### Fetal and neonatal outcomes

Low-certainty evidence suggests that intermittent iron supplementation may have a similar effect to daily iron supplementation on low birth weight (< 2500 g) (8 trials, 1898 neonates; RR: 0.82, 95% CI: 0.50–1.22). However, the evidence on preterm birth and very preterm birth was assessed as very uncertain. Evidence on the relative effects of intermittent versus daily iron supplementation on neonatal mortality is also very uncertain. Neonatal infections and SGA outcomes were not included in the review.

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

Intermittent iron and folic acid supplementation might cost a little less than daily iron and folic acid supplementation due to the lower total weekly dose of iron.

### Equity

Intermittent iron and folic acid supplementation may have less impact on health inequalities than daily iron and folic acid supplementation, as anaemia is more common in disadvantaged populations.

### Acceptability

Qualitative evidence suggests that the availability of iron supplements may actively encourage women to engage with ANC providers (low confidence in the evidence) (22). However, where there are additional costs associated with supplementation or where the supplements may be unavailable (because of resource constraints) women are less likely to engage with ANC services (high confidence in the evidence). Women may find intermittent iron supplementation more acceptable than daily iron supplementation,

particularly if they experience side-effects with daily iron supplements.

### Feasibility

Intermittent iron may be more feasible in some low-resource settings if it costs less than daily iron.

## A.3: Calcium supplements

**RECOMMENDATION A.3: In populations with low dietary calcium intake, daily calcium supplementation (1.5–2.0 g oral elemental calcium) is recommended for pregnant women to reduce the risk of pre-eclampsia.** (*Context-specific recommendation*)

### Remarks

- This recommendation is consistent with the 2011 *WHO recommendations for prevention and treatment of pre-eclampsia and eclampsia* (57) (strong recommendation, moderate-quality evidence) and supersedes the WHO recommendation found in the 2013 *Guideline: calcium supplementation in pregnant women* (38).
- Dietary counselling of pregnant women should promote adequate calcium intake through locally available, calcium-rich foods.
- Dividing the dose of calcium may improve acceptability. The suggested scheme for calcium supplementation is 1.5–2 g daily, with the total dose divided into three doses, preferably taken at mealtimes.
- Negative interactions between iron and calcium supplements may occur. Therefore, the two nutrients should preferably be administered several hours apart rather than concomitantly (38).
- As there is no clear evidence on the timing of initiation of calcium supplementation, stakeholders may wish to commence supplementation at the first ANC visit, given the possibility of compliance issues.
- To reach the most vulnerable populations and ensure a timely and continuous supply of supplements, stakeholders may wish to consider task shifting the provision of calcium supplementation in community settings with poor access to health-care professionals (see Recommendation E.6.1, in section E: Health systems interventions to improve the utilization and quality of ANC).
- The implementation and impact of this recommendation should be monitored at the health service, regional and country levels, based on clearly defined criteria and indicators associated with locally agreed targets. Successes and failures should be evaluated to inform integration of this recommendation into the ANC package.
- Further WHO guidance on prevention and treatment of pre-eclampsia and eclampsia is available in the 2011 WHO recommendations (57), available at: [http://apps.who.int/iris/bitstream/10665/44703/1/9789241548335\\_eng.pdf](http://apps.who.int/iris/bitstream/10665/44703/1/9789241548335_eng.pdf)

## Summary of evidence and considerations

### Effects of calcium supplements compared with no calcium supplements (for outcomes other than hypertension/pre-eclampsia) (EB Table A.3)

Evidence on the effects of calcium supplements on outcomes other than hypertension/pre-eclampsia was derived from a Cochrane systematic review (58). The review included data from 23 trials involving 18 587 pregnant women. The aim of the review was to determine the effect of calcium on maternal and perinatal outcomes other than hypertension. There is a separate Cochrane review on the latter (59), which has been referenced to support existing WHO

recommendations on calcium supplementation to prevent pre-eclampsia in populations with low dietary calcium intake (38, 57).

In 14 trials, daily calcium doses ranged from 1000 mg to 2000 mg, and in the remainder it was less than 1000 mg. Eleven trials started calcium supplementation at or after 20 weeks of gestation, five trials started before 20 weeks, and the rest did not specify when supplementation was initiated. The primary outcome of 16 of the trials was pregnancy-induced hypertension. For outcomes other than hypertension, few trials contributed to each outcome; this is the evidence presented in this section.

## Maternal outcomes

High-certainty evidence shows that calcium supplementation does not have important effects on maternal anaemia (1 trial, 1098 women; RR: 1.04, 95% CI: 0.90–1.22) or caesarean section rates (9 trials, 7440 women; RR: 0.99, 95% CI: 0.89–1.10). Moderate-certainty evidence indicates that calcium supplementation probably has little or no effect on maternal mortality (2 trials, 8974 women; RR: 0.29, 95% CI: 0.06–1.38) and probably makes little or no difference to the risk of urinary tract infections (3 trials, 1743 women; RR: 0.95, 95% CI: 0.69–1.30). Low-certainty evidence suggests that calcium supplementation may make little or no difference to maternal weight gain (3 trials; MD: –29.46 g per week, 95% CI: –119.80 to 60.89 g per week). Maternal satisfaction was not reported in any of the trials included in the Cochrane review.

**Side-effects:** Calcium supplementation makes little or no difference to the risk of “any side-effect”, a composite outcome including headache, vomiting, backache, swelling, vaginal and urinary complaints, dyspepsia and abdominal pain (1 trial, 8312 women; RR: 1.02, 95% CI: 0.93–1.12), and probably makes little or no difference to the risk of urinary stones (3 trials, 13 419 women; RR: 1.11, 95% CI: 0.48–2.54), renal colic (1 trial, 8312 women; RR: 1.67, 95% CI: 0.40–6.99) and impaired renal function (1 trial, 4589 women; RR: 0.91, 95% CI: 0.51–1.64), all assessed as moderate-certainty evidence. Low-certainty evidence suggests that it may have little or no effect on the risk of gallstones (1 trial, 518 women; RR: 1.35, 95% CI: 0.48–3.85).

## Fetal and neonatal outcomes

Calcium supplementation probably has little or no effect on low-birth-weight babies (< 2500 g), as indicated by evidence that was of moderate certainty due to inconsistency (6 trials, 14 162 women; RR: 0.93, 95% CI: 0.81–1.07). Low-certainty evidence suggests that it may have little or no effect on preterm birth before 37 weeks of gestation (13 trials, 16 139 women; RR: 0.86, 95% CI: 0.70–1.05). However, when trials are stratified by dose (< 1000 mg vs ≥ 1000 mg), moderate-certainty evidence shows that high-dose calcium supplementation probably reduces preterm birth (12 trials, 15 479 women; RR: 0.81, 95% CI: 0.66–0.99).

Low-certainty evidence suggests that calcium supplementation may make little or no difference to perinatal mortality (8 trials, 15 785 women; RR: 0.87, 95% CI: 0.72–1.06), and moderate-certainty evidence shows that it probably has little or no effect

on stillbirths or fetal deaths (6 trials, 15 269 women; RR: 0.91, 95% CI: 0.72–1.14).

## Additional considerations

- In the *WHO recommendations for prevention and treatment of pre-eclampsia and eclampsia* (2011), the recommendation on calcium states: “In areas where dietary calcium intake is low, calcium supplementation during pregnancy (at doses of 1.5–2.0 g elemental calcium/day) is recommended for the prevention of pre-eclampsia in all women, but especially in those at high risk of developing pre-eclampsia (strong recommendation)” (57). This recommendation is based on moderate-quality evidence showing a 64% risk reduction (CI: 35–80%) in pre-eclampsia among women or populations with low baseline dietary calcium intake (57).
- In considering the evidence from the review of “non-hypertensive” effects, the GDG agreed that the effect of calcium on preterm birth is probably not distinct from the effect on preventing pre-eclampsia, as preterm birth is frequently a consequence of pre-eclampsia.

## Values

Please see “Women’s values” in section 3.A:

Background (p. 15).

## Resources

The GDG noted that the cost of calcium (3 × 1 tablet 600 mg per day for 6 months = US\$ 11.50) (27) is relatively high compared with supplements such as iron and folic acid. The weight of the supplement may also have cost and logistical implications with respect to storage and transport.

## Equity

In many LMICs, women who are poor, least educated and residing in rural areas have worse pregnancy outcomes than do more advantaged women (29). Preterm birth is the most common cause of neonatal mortality, with the majority of deaths occurring in LMICs. Therefore, effective nutritional interventions in disadvantaged populations aimed at reducing preterm birth could help to address health inequalities.

## Acceptability

Qualitative evidence indicates that women in a variety of settings tend to view ANC as a source of knowledge and information and that they generally appreciate any advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22). However,



calcium carbonate tablets might be unpalatable to many women, as they can be large and have a powdery texture (59). In addition, this intervention usually involves taking three tablets a day, which significantly increasing the number of tablets a woman is required to take on a daily basis (i.e. in addition to iron and folic acid). This could have implications for both acceptability and compliance, which needs to be assessed in a programmatic context.

### Feasibility

In addition to the cost, providing calcium supplements may be associated with logistical issues (e.g. supplements are bulky and require adequate transport and storage to maintain stock in facilities) and other challenges (e.g. forecasting). Qualitative evidence on health-care providers' views suggests that resource constraints may limit implementation (high confidence in the evidence) (45).

## A.4: Vitamin A supplements

**RECOMMENDATION A.4: Vitamin A supplementation is only recommended for pregnant women in areas where vitamin A deficiency is a severe public health problem, to prevent night blindness.** (*Context-specific recommendation*)

### Remarks

- This recommendation supersedes the previous WHO recommendation found in the 2011 *Guideline: vitamin A supplementation in pregnant women* (60).
- Vitamin A is not recommended to improve maternal and perinatal outcomes.
- Vitamin A deficiency is a severe public health problem if 5% or more of women in a population have a history of night blindness in their most recent pregnancy in the previous 3–5 years that ended in a live birth, or if 20% or more of pregnant women have a serum retinol level below 0.70 µmol/L (61). Determination of vitamin A deficiency as a public health problem involves estimating the prevalence of deficiency in a population by using specific biochemical and clinical indicators of vitamin A status.
- Pregnant women should be encouraged to receive adequate nutrition, which is best achieved through consumption of a healthy, balanced diet, and to refer to WHO guidance on healthy eating (41).
- In areas where supplementation is indicated for vitamin A deficiency, it can be given daily or weekly. Existing WHO guidance suggests a dose of up to 10 000 IU vitamin A per day, or a weekly dose of up to 25 000 IU (60).
- A single dose of a vitamin A supplement greater than 25 000 IU is not recommended as its safety is uncertain. Furthermore, a single dose of a vitamin A supplement greater than 25 000 IU might be teratogenic if consumed between day 15 and day 60 from conception (60).
- There is no demonstrated benefit from taking vitamin A supplements in populations where habitual daily vitamin A intakes exceed 8000 IU or 2400 µg, and the potential risk of adverse events increases with higher intakes (above 10 000 IU) if supplements are routinely taken by people in these populations (62).

## Summary of evidence and considerations

### Effects of vitamin A supplements compared with no vitamin A supplements (EB Table A.4)

The evidence was derived from a Cochrane systematic review of 19 trials of vitamin A (with or without other supplements) compared with no vitamin A (or placebo, or other supplements) involving over 310 000 women (63). All but one trial (conducted in the United Kingdom) were conducted in LMICs, including Bangladesh, China, Ghana, India, Indonesia, Malawi, Nepal, South Africa and the United Republic of Tanzania. Most trials were

conducted in vitamin A deficient populations, with one study including only women living with HIV. Trials varied considerably in design, including in the dose and timing of the intervention. Ten trials contributed data to the comparison of vitamin A alone versus placebo or no treatment.

### Maternal outcomes

Moderate-certainty evidence shows that vitamin A supplementation in vitamin A deficient populations during pregnancy probably reduces maternal anaemia (3 trials, 15 649 women; RR: 0.64, 95% CI: 0.43–0.94), but that it probably has little or no effect on



maternal mortality (4 trials, 101 574 women; RR: 0.88, 95% CI: 0.65–1.20). Low-certainty evidence on a composite outcome for maternal infection (including fever for more than one week at one week postnatally, puerperal fever greater than 38°C, subclinical mastitis and/or bacterial vaginosis) suggests that vitamin A supplementation may reduce maternal infection (5 trials, 17 313 women; average RR: 0.45, 95% CI: 0.2–0.99). Side-effects and other maternal ANC guideline outcomes were not reported in the trials.

### Fetal and neonatal outcomes

High-certainty evidence shows that vitamin A supplementation makes little or no difference to perinatal mortality (76 176 women; RR: 1.01, 95% CI: 0.95–1.07), neonatal mortality (3 trials, 89 556 neonates; RR: 0.97, 95% CI: 0.90–1.05) or stillbirths (2 trials, 122 850 neonates; RR: 1.04, 95% CI: 0.98–1.10). Moderate-certainty evidence indicates that vitamin A supplementation probably has little or no effect on low birth weight (< 2500 g) (4 trials, 14 599 neonates; RR: 0.102, 95% CI: 0.89–1.16), and low-certainty evidence suggests that it may have little or no effect on preterm birth (5 trials, 40 137 women; RR: 0.98, 95% CI: 0.94–1.01). Neonatal infections and congenital anomalies were not reported in the trials.

### Additional considerations

- Moderate-certainty evidence shows that vitamin A supplementation reduces night blindness in pregnant women living in areas with a high prevalence of this condition (2 trials, approximately 100 000 women; RR: 0.79, 95% CI: 0.64–0.98).
- Miscarriage and teratogenicity have been associated with high vitamin A intake within 60 days of conception; however, a WHO expert group consultation in 1998 concluded that daily doses of

up to 3000 µg per day after day 60 are probably safe, especially in areas where vitamin A deficiency is common (62).

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

Vitamin A supplements are relatively inexpensive at approximately US\$ 0.30 per woman per month (10 000 IU per day or 25 000 IU per week) (27).

Vitamin A can be given as a daily or weekly supplement.

### Equity

Effective nutritional interventions in disadvantaged populations could help to address health inequalities by improving nutritional status and promoting good maternal health.

### Acceptability

Qualitative evidence suggests that women in a variety of settings tend to view ANC as a source of knowledge and information and that they generally appreciate any advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22).

### Feasibility

Qualitative evidence shows that where there are additional costs associated with supplements (high confidence in the evidence) or where the recommended intervention is unavailable because of resource constraints (low confidence in the evidence), women may be less likely to engage with ANC (45).

## A.5: Zinc supplements

**RECOMMENDATION A.5: Zinc supplementation for pregnant women is only recommended in the context of rigorous research.** (*Context-specific recommendation – research*)

### Remarks

- Many of the included studies were at risk of bias, which influenced the certainty of the review evidence on the effects of zinc supplementation.
- The low-certainty evidence that zinc supplementation may reduce preterm birth warrants further investigation, as do the other outcomes for which the evidence is very uncertain (e.g. perinatal mortality, neonatal sepsis), particularly in zinc-deficient populations with no food fortification strategy in place. Further research should aim to clarify to what extent zinc supplementation competes with iron and/or calcium antenatal supplements for absorption. The GDG considered that food fortification may be a more cost-effective strategy and that more evidence is needed on the cost-effectiveness of food fortification strategies.

## Summary of evidence and considerations

### Effects of zinc supplements compared with no zinc supplements (EB Table A.5)

The evidence was derived from a Cochrane review that included 21 trials involving more than 17 000 women (64). Most studies were conducted in LMICs, including Bangladesh, Chile, China, Egypt, Ghana, Indonesia, the Islamic Republic of Iran, Nepal, Pakistan, Peru and South Africa. Six trials were conducted in Denmark, the United Kingdom and the USA. Daily zinc supplementation was compared with no intervention or placebo. There was a wide variation among trials in terms of trial size (range: 56–4926 women), zinc dosage (range: 5–90 mg per day), nutritional and zinc status at trial entry, initiation and duration of supplementation (starting before conception in one trial, first or second trimester in the majority, or after 26 weeks of gestation in two trials, until delivery), and compliance with treatment.

### Maternal outcomes

Moderate-certainty evidence indicates that zinc supplementation probably makes little or no difference to the risk of any maternal infections (3 trials, 1185 women; RR: 1.06; 95% CI: 0.74–1.53). The evidence on caesarean section, pre-eclampsia and side-effects (maternal taste and smell dysfunction) is of very low certainty, and the review did not include anaemia, maternal mortality or maternal satisfaction as review outcomes.

### Fetal and neonatal outcomes

Moderate-certainty evidence indicates that zinc supplementation probably makes little or no difference to the risk of having SGA (8 trials, 4252

newborns; RR: 1.02; 95% CI: 0.94–1.11) or low-birth-weight neonates (14 trials, 5643 neonates; RR: 0.93, 95% CI: 0.78–1.12). However, low-certainty evidence suggests that zinc supplementation may reduce preterm birth (16 trials, 7637 women; RR: 0.86, 95% CI: 0.76–0.97), particularly in women with presumed low zinc intake or poor nutrition (14 trials, 7099 women; RR: 0.87, 95% CI: 0.77–0.98).

Low-certainty evidence suggests that zinc supplementation may have little or no effect on congenital anomalies (6 trials, 1240 newborns; RR: 0.67, 95% CI: 0.33–1.34) and macrosomia (defined in the review as “high birth weight”; 5 trials, 2837 neonates; RR: 1.00, 95% CI: 0.84–1.18). Evidence on perinatal mortality and neonatal sepsis is of very low certainty.

### Additional considerations

- The trials were clinically heterogeneous, therefore it is unclear what dose and timing of zinc supplementation, if any, might lead to a possible reduction in preterm birth.
- There is little or no evidence on side-effects of zinc supplementation. In addition, it is unclear to what extent zinc might compete with iron and/or calcium for absorption. Maternal anaemia was not evaluated in the review.

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

Zinc costs approximately US\$ 1.30 for 100 tablets of 20 mg (i.e. less than US\$ 3.00 for a 6-month supply based on a daily dose of 20 mg) (27).

### Equity

Effective interventions to improve maternal nutrition in disadvantaged populations could help to address health inequalities. A WHO report shows that inequalities in neonatal, infant and child mortality, as well as stunting prevalence, can be demonstrated according to economic status, education and place of residence in LMICs. The prevalence of stunting may be a good indicator of zinc deficiency in LMICs (39).

### Acceptability

Qualitative evidence suggests that women in a variety of settings tend to view ANC as a source

of knowledge and information and they generally appreciate any professional advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22).

### Feasibility

It may be more feasible to fortify food with zinc rather than to provide zinc as a single supplement, particularly in settings with a high prevalence of stunting in children.

## A.6: Multiple micronutrient (MMN) supplements

**RECOMMENDATION A.6: Multiple micronutrient supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes.** (*Not recommended*)

### Remarks

- There is some evidence of additional benefit of MMN supplements containing 13–15 different micronutrients (including iron and folic acid) over iron and folic acid supplements alone, but there is also some evidence of risk, and some important gaps in the evidence. Although the GDG agreed that overall there was insufficient evidence to warrant a recommendation, the group agreed that policy-makers in populations with a high prevalence of nutritional deficiencies might consider the benefits of MMN supplements on maternal health to outweigh the disadvantages, and may choose to give MMN supplements that include iron and folic acid.
- More research is needed to determine which micronutrients improve maternal and perinatal outcomes, and how these can be optimally combined into a single supplement.

### Summary of evidence and considerations

#### Effects of MMN supplements (with 13–15 different MMNs) compared with iron and folic acid supplements (EB Table A.6)

The evidence was derived from a Cochrane review that included 17 trials involving 137 791 women (65); however, only 14 trials contributed data to this comparison. These 14 trials were all conducted in LMICs: Bangladesh (2), Burkina Faso (1), China (2), Guinea-Bissau (1), Indonesia (2), Mexico (1), Nepal (2), Niger (1), Pakistan (1) and Zimbabwe (1). The trials compared supplements containing 13–15 micronutrients (including iron and folic acid) with iron and folic acid supplements only, except for one trial in which the control arm comprised iron only. Nine trials evaluated supplements with 15 micronutrients, including vitamin A, B1, B2, B6, B12, C, D and E, copper, folic acid, iodine, iron, niacin, selenium and zinc, with exactly the same dosages as the UN international MMN preparation (UNIMMAP

(66). Evidence from these UNIMMAP trials was synthesized together with trials of 13 and 14 MMN supplements, and in separate subgroup analyses using the random effects method. Subgroup analyses were performed according to the dose of iron (60 mg or 30 mg) used in the control arm. Analyses can be found in the Web supplement (EB Table A.6).

#### Maternal outcomes

High-certainty evidence shows that MMN supplementation has a similar effect to iron and folic acid supplements only (standard care) on maternal anaemia (5 trials; RR: 0.98, 95% CI: 0.85–1.13). Compared to iron and folic acid only, moderate-certainty evidence indicates that MMN supplements probably make little or no difference to caesarean section rates (4 trials; RR: 1.03, 95% CI: 0.75–1.43) and low-certainty evidence suggests that they may have little or no effect on maternal mortality (3 trials; RR: 0.97, 95% CI: 0.63–1.48). There was no evidence relating to maternal satisfaction or side-effects.

### Fetal and neonatal outcomes

High-certainty evidence shows that MMN supplementation reduces the risk of having a low-birth-weight neonate compared with iron and folic acid supplements only (14 trials; RR: 0.88, 95% CI: 0.85–0.91), but moderate-certainty evidence indicates that it probably makes little or no difference to the risk of having an SGA neonate (13 trials; RR: 0.98, 95% CI: 0.96–1.00). High-certainty evidence shows that MMN supplements make little or no difference to preterm birth rates (14 trials; RR: 0.95, 95% CI: 0.88–1.03). Moderate-certainty evidence shows that MMN supplements probably make little or no difference to perinatal mortality (11 trials; RR: 1.00, 95% CI: 0.85–1.19), neonatal mortality (11 trials; RR: 0.99, 95% CI: 0.90–1.08) or stillbirths (14 trials; RR: 0.97, 95% CI: 0.86–1.09). The evidence on congenital anomalies is of low certainty and inconclusive (1 trial, 1200 women; RR: 0.99, 95% CI: 0.14–7.00).

High-certainty evidence from analyses restricted to trials of UNIMMAP only are consistent with the overall findings, with the exception that it shows that UNIMMAP reduces the risk of having an SGA neonate compared with iron and folic acid supplements only (8 trials; RR: 0.85, 95% CI: 0.77–0.94).

Subgroup analyses according to the iron dose in the control group are generally consistent with the overall findings. However, for the subgroup of studies that compared MMN supplements to 60 mg iron and 400 µg folic acid, a harmful effect of MMNs on neonatal mortality cannot be excluded (6 trials; RR: 1.22, 95% CI: 0.95–1.57).

### Additional considerations

- A separate review of the effects of MMN supplementation during pregnancy on child health benefits pooled data from nine of the trials included in the Cochrane review and found no evidence of beneficial effects on child mortality, growth or cognitive function (67).

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

UNIMMAP supplements cost about US\$ 3 per woman per pregnancy, whereas iron and folic acid supplementation costs less than US\$ 1 (27).

### Equity

Effective interventions to improve maternal nutrition in disadvantaged populations could help to address maternal and neonatal health inequalities by improving maternal health and preventing illness related to nutritional deficiencies. However, the cost difference between MMNs and iron and folic acid supplementation may have an impact on affordability for disadvantaged populations, especially those in remote and rural areas, because they are often expected to pay for visits and supplements in addition to bearing greater transport costs due to the greater distance to travel to ANC services (68).

### Acceptability

Qualitative evidence suggests that women in a variety of settings tend to view ANC as a source of knowledge and information and that they generally appreciate any advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22). However, it has been noted that the lack of appropriate training on MMN supplementation has been reported by health-care providers as a major gap (68).

### Feasibility

From the demand side, MMN supplementation should be as feasible as iron and folic acid supplementation if supplements are free and available, and it will face the same challenges in terms of compliance. However, on the supply side, there may be several barriers to overcome, such as changes in regulatory norms and policies (e.g. tariffs, labelling, imports, government oversight, etc.), ensuring sustainable MMN production (local or imported), product availability and quality. Great variability in feasibility across countries and within them would be expected (68).

## A.7: Vitamin B6 (pyridoxine) supplements

**RECOMMENDATION A.7: Vitamin B6 (pyridoxine) supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes. (Not recommended)**

### Remarks

- Pregnant women should be encouraged to receive adequate nutrition, which is best achieved through consumption of a healthy, balanced diet, and to refer to guidelines on healthy eating (41).
- The GDG agreed that there is insufficient evidence on the benefits and harms, if any, of routine vitamin B6 supplementation in pregnancy. However, research on the effects of routine vitamin B6 supplementation for pregnant women on maternal and perinatal outcomes is not considered a research priority.

### Summary of evidence and considerations

#### Effects of vitamin B6 supplements compared with no vitamin B6 supplements (EB Table A.7)

The evidence was derived from a Cochrane review that included four trials involving approximately 1646 pregnant women (69). Studies were conducted in HICs between 1960 and 1984. Vitamin B6 (pyridoxine) given intramuscularly as a single dose (100 mg) or orally as capsules or lozenges (2.6 mg to 20 mg per day) was compared with placebo or no treatment. Only two out of four studies contributed data to this comparison.

#### Maternal outcomes

Low-certainty evidence suggests that oral pyridoxine supplements may have little or no effect on pre-eclampsia (2 trials, 1197 women; RR: 1.71, 95% CI: 0.85–3.45). No other maternal outcomes relevant to the ANC guideline were reported in the review.

#### Fetal and neonatal outcomes

Trials contributed no data on low birth weight, preterm birth or other ANC guideline outcomes. Mean birth weight was evaluated in one small trial but the evidence is very uncertain. There was no evidence on congenital anomalies.

#### Additional considerations

- Moderate-certainty evidence shows that vitamin B6 probably provides some relief for nausea during pregnancy (see evidence summary for Recommendation D.1, in section D: Interventions for common physiological symptoms).
- Vitamin B6 deficiency alone is uncommon; it mostly occurs in combination with deficiencies of other B vitamins (70).

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

As a single supplement, vitamin B6 (pyridoxine hydrochloride tablets) can cost about US\$ 2.50 for 90 × 10 mg tablets (71).

### Equity

Effective interventions to improve maternal nutrition in disadvantaged populations could help to address health inequalities.

### Acceptability

Qualitative evidence suggests that women in a variety of settings tend to view ANC as a source of knowledge and information and that they generally appreciate any professional advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22).

### Feasibility

Qualitative evidence shows that where there are additional costs associated with supplements (high confidence in the evidence) or where the recommended intervention is unavailable because of resource constraints (low confidence in the evidence), women may be less likely to engage with ANC services (45).

## A.8: Vitamin E and C supplements

**RECOMMENDATION A.8: Vitamin E and C supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes.** (*Not recommended*)

### Remarks

- The GDG noted that vitamin E and C combined supplements were evaluated mainly in the context of preventing pre-eclampsia. Vitamin C is important for improving the bioavailability of oral iron, but this was not considered within the context of the Cochrane reviews. In addition, low-certainty evidence on vitamin C alone suggests that it may prevent prelabour rupture of membranes (PROM). Therefore, the GDG agreed that future research should consider vitamin C supplements separately from vitamin E and C supplements.
- Pregnant women should be encouraged to receive adequate nutrition, which is best achieved through consumption of a healthy, balanced diet, and to refer to guidelines on healthy eating (41). It is relatively easy to consume sufficient quantities of vitamin C from food sources.

## Summary of evidence and considerations

### Effects of vitamin E and C supplements compared with no vitamin E and C supplements (EB Table A.8)

The evidence was derived from two Cochrane systematic reviews that included 17 trials conducted in low-, middle- and high-income countries contributed data (72, 73). The trials assessed vitamin E plus vitamin C combined supplements compared with placebo or no vitamin E and C supplements. The most commonly used dose of vitamin E was 400 IU daily (15 trials) and vitamin C was 1000 mg daily (13 trials). The primary outcome of 14 trials was pre-eclampsia and nine of the trials recruited women at “high” or “increased” risk of pre-eclampsia. Most of the trials commenced supplementation in the second trimester.

### Maternal outcomes

Moderate-certainty evidence shows that vitamin E and C combined supplements probably have little or no effect on the risk of developing pre-eclampsia (14 studies, 20 878 women; RR: 0.91 95% CI: 0.79–1.06) and eclampsia (8 trials, 19 471 women; RR: 1.67, 95% CI: 0.82–3.41). Moderate-certainty evidence also shows that vitamin E and C supplements probably have little or no effect on maternal mortality (7 trials, 17 120 women; RR: 0.60, 95% CI: 0.14–2.51) and caesarean section (6 trials, 15 297 women; RR: 1.02, 95% CI: 0.97–1.07).

**Side-effects:** High-certainty evidence shows that vitamin E and C supplementation is associated with an increased risk of abdominal pain during pregnancy (1 trial, 1877 women; RR: 1.66, 95% CI: 1.16–2.37; absolute effect of 32 more per 1000 women).

### Fetal and neonatal outcomes

High-certainty evidence indicates that vitamin E and C supplementation does not have an important effect on SGA (11 trials, 20 202 women; RR: 0.98, 95% CI: 0.91–1.06). Moderate-certainty evidence shows that vitamin E and C supplements probably have little or no effect on preterm birth (11 trials, 20 565 neonates; RR: 0.98, 95% CI: 0.88–1.09), neonatal infections (5 trials, 13 324 neonates; RR: 1.10, 95% CI: 0.73–1.67) and congenital anomalies (4 trials, 5511 neonates; RR: 1.16, 95% CI: 0.83–1.63).

### Additional considerations

- The high-certainty evidence on abdominal pain is derived from a large, well designed trial in which abdominal pain occurred in 7.9% of women in the vitamin E and C supplement group and 4.8% of women in the placebo group.
- Despite the certainty of these effects of vitamin E and C supplementation, the biological explanations for these adverse effects are not established.
- Moderate-certainty evidence indicates that vitamin E and C supplements probably reduce the risk of placental abruption (7 trials, 14 922 women; RR: 0.64, 95% CI: 0.44–0.93; absolute effect of 3 fewer abruptions per 1000) but make little or no difference to the risk of antepartum haemorrhage from any cause (2 trials, 12 256 women; RR: 1.25, 95% CI: 0.85–1.82).
- High-certainty evidence shows vitamin E and C supplementation increases PROM at term (37 weeks of gestation or more) (2 trials, 2504 women; RR: 1.77, 95% CI: 1.37–2.28; absolute effect of 52 more cases of PROM per 1000).
- The trial contributing the most data on PROM was stopped early, based on their PROM data, when



only a quarter of the planned sample (10 000 women) had been accrued.

- Low- to moderate-certainty evidence on vitamin C only suggests that vitamin C alone (in doses ranging from 100 mg to 1000 mg) may reduce preterm PROM (5 studies, 1282 women; RR: 0.66, 95% CI: 0.48–0.91) and term PROM (1 study, 170 women; RR: 0.55, 95% CI: 0.32–0.94).

### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

### Resources

Vitamin E (tocopherol) 400 IU daily can cost about US\$ 8 for a month’s supply. Costs of vitamin C vary widely; chewable vitamin C tablets (1000 mg) can cost about US\$ 3 for a month’s supply (74).

### Equity

Effective interventions to reduce pre-eclampsia could help to address health inequalities because

mortality from pre-eclampsia mainly occurs among disadvantaged populations.

### Acceptability

Qualitative evidence suggests that women in a variety of settings tend to view ANC as a source of knowledge and information and that they generally appreciate any professional advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22).

### Feasibility

Qualitative evidence shows that where there are additional costs associated with supplements (high confidence in the evidence) or where the recommended intervention is unavailable because of resource constraints (low confidence in the evidence), women may be less likely to engage with ANC services (45).

## A.9: Vitamin D supplements

**RECOMMENDATION A.9: Vitamin D supplementation is not recommended for pregnant women to improve maternal and perinatal outcomes.** (*Not recommended*)

### Remarks

- This recommendation supersedes the previous WHO recommendation found in the 2012 Guideline: vitamin D supplementation in pregnant women (75).
- Pregnant women should be advised that sunlight is the most important source of vitamin D. The amount of time needed in the sun is not known and depends on many variables, such as the amount of skin exposed, the time of day, latitude and season, skin pigmentation (darker skin pigments synthesize less vitamin D than lighter pigments) and sunscreen use (75).
- Pregnant women should be encouraged to receive adequate nutrition, which is best achieved through consumption of a healthy, balanced diet, and to refer to guidelines on healthy eating (47).
- For pregnant women with documented vitamin D deficiency, vitamin D supplements may be given at the current recommended nutrient intake (RNI) of 200 IU (5 µg) per day.
- According to the Cochrane review, there are 23 ongoing or unpublished studies on vitamin D supplementation in pregnancy (76). Evidence from these trials should help to clarify the current uncertainties regarding vitamin D effects, particularly the effect on preterm birth, and any other associated benefits or harms of vitamin D when combined with other vitamins and minerals, particularly calcium.

### Summary of evidence and considerations

The evidence was derived from a Cochrane systematic review that included 15 trials assessing 2833 women (76). Nine trials were conducted in LMICs (Bangladesh, Brazil, China, India and the

Islamic Republic of Iran) and six were conducted in HICs (France, New Zealand, Russia and the United Kingdom). Sample sizes ranged from 40 to 400 women. Nine trials compared the effects of vitamin D alone versus placebo or no supplementation, and six trials compared the effects of vitamin D plus calcium

versus placebo or no supplementation. The dose and regimen of vitamin D varied widely among the trials.

#### a) Effects of vitamin D supplements alone versus placebo or no supplement (EB Table A.9)

Nine trials contributed data to this comparison. Six trials evaluated daily vitamin D with daily doses ranging from 400 IU to 2000 IU. Two trials evaluated a single dose of 200 000 IU given at about 28 weeks of gestation, one trial evaluated a weekly dose of 35 000 IU during the third trimester, and one trial administered 1–4 vitamin D doses (60 000–480 000 IU in total) depending on the participants' baseline serum 25-hydroxy-vitamin D levels.

##### Maternal outcomes

The evidence on pre-eclampsia, GDM, maternal mortality, caesarean section and side-effects is very uncertain (i.e. all findings were assessed as very low-certainty evidence).

##### Fetal and neonatal outcomes

Low-certainty evidence suggests that vitamin D supplementation may reduce low-birth-weight neonates (3 trials, 493 women; RR: 0.40, 95% CI: 0.24–0.67) and preterm birth (< 37 weeks of gestation) (3 trials, 477 women; RR: 0.36, 95% CI: 0.14–0.93), but may have little or no effect on neonatal deaths (2 trials, 282 women, RR: 0.27; 95% CI: 0.04–1.67) and stillbirths (3 trials, 540 women; RR: 0.35, 95% CI: 0.06–1.99).

#### b) Effects of vitamin D plus calcium supplements versus placebo or no supplement (EB Table A.9)

Six trials contributed data to this comparison. Vitamin D doses ranged from 200 IU to 1250 IU daily and calcium doses ranged from 375 mg to 1250 mg daily.

##### Maternal outcomes

Moderate-certainty evidence shows that vitamin D plus calcium probably reduces pre-eclampsia (3 trials, 798 women; RR: 0.51; 95% CI: 0.32–0.80), but low-certainty evidence suggest that it may have little or no effect on GDM (1 trial, 54 women, 1 event; RR: 0.43, 95% CI: 0.05–3.45).

##### Fetal and neonatal outcomes

Moderate-certainty evidence indicates that vitamin D plus calcium probably increases preterm birth

(< 37 weeks of gestation) (3 trials, 798 women; RR: 1.57, 95% CI: 1.02–2.43). Low-certainty evidence suggests that vitamin D plus calcium has little or no effect on neonatal mortality (1 trial, 660 women; RR: 0.20, 95% CI: 0.01–4.14).

##### Additional considerations

- Due to the limited evidence currently available to directly assess the benefits and harms of the use of vitamin D supplementation alone in pregnancy for improving maternal and infant health outcomes, the use of this intervention during pregnancy as part of routine ANC is not recommended (75).
- The moderate-certainty evidence showing that adding vitamin D to calcium supplementation probably increases preterm birth is of concern and this potential harm needs further investigation.

##### Values

Please see “Women’s values” in section 3.A: Background (p. 15).

##### Resources

Vitamin D supplements can cost from US\$ 2 per month, depending on the dose prescribed (74).

##### Equity

Effective interventions to improve maternal nutrition in disadvantaged populations could help to address health inequalities.

##### Acceptability

Qualitative evidence suggests that women in a variety of settings tend to view ANC as a source of knowledge and information and that they generally appreciate any professional advice (including dietary or nutritional) that may lead to a healthy baby and a positive pregnancy experience (high confidence in the evidence) (22).

##### Feasibility

Qualitative evidence shows that where there are additional costs associated with supplements (high confidence in the evidence) or where the recommended intervention is unavailable because of resource constraints (low confidence in the evidence), women may be less likely to engage with ANC services (45).

## A.10: Restricting caffeine intake

**RECOMMENDATION A.10: For pregnant women with high daily caffeine intake (more than 300 mg per day),<sup>a</sup> lowering daily caffeine intake during pregnancy is recommended to reduce the risk of pregnancy loss and low-birth-weight neonates. (Context-specific recommendation)**

### Remarks

- Pregnant women should be informed that a high daily caffeine intake (> 300 mg per day) is probably associated with a higher risk of pregnancy loss and low birth weight.
- Caffeine is a stimulant found in tea, coffee, soft-drinks, chocolate, kola nuts and some over-the-counter medicines. Coffee is probably the most common source of high caffeine intake. A cup of instant coffee can contain about 60 mg of caffeine; however, some commercially brewed coffee brands contain more than 150 mg of caffeine per serving.
- Caffeine-containing teas (black tea and green tea) and soft drinks (colas and iced tea) usually contain less than 50 mg per 250 mL serving.

a This includes any product, beverage or food containing caffeine (i.e. brewed coffee, tea, cola-type soft drinks, caffeinated energy drinks, chocolate, caffeine tablets).

### Summary of evidence and considerations

#### a) Effects of decaffeinated coffee versus caffeinated coffee (RCT evidence) (EB Table A.10a)

Some evidence on the effects of caffeine intake was derived from a Cochrane review that included two RCTs (40). Only one of the trials, conducted in Denmark, contributed evidence. In this trial, 1207 pregnant women drinking more than three cups of coffee a day were randomized to receive instant decaffeinated coffee (intervention group) versus instant caffeinated coffee (control group) in order to assess the effect of caffeine reduction during pregnancy. In this trial, a cup of caffeinated coffee was estimated to contain 65 mg caffeine. Other sources of caffeine, such as cola, tea and chocolate were not restricted. Mean daily caffeine intake in the decaffeinated coffee group was 117 mg per day (interquartile range [IQR]: 56–228 mg) compared with 317 mg per day (IQR: 229–461 mg) in the caffeinated coffee group.

#### Maternal outcomes

None of the maternal outcomes addressed in the ANC guideline were reported in the review.

#### Fetal and neonatal outcomes

Low-certainty evidence from one trial shows that restricting caffeine intake (replacing caffeinated coffee with decaffeinated coffee) may have little or no effect on SGA (1150 neonates; RR: 0.97, 95% CI: 0.57–1.64), mean birth weight (1197 neonates; MD: 20.00, 95% CI: –48.68 to 88.68) and preterm birth (1153 neonates; RR: 0.81, 95% CI: 0.48–1.37).

No data were available on congenital anomalies or perinatal mortality.

#### b) Effects of high caffeine intake versus moderate, low or no caffeine intake (non-randomized study evidence) (EB Table A.10b)

The GDG considered the evidence from RCTs to be insufficient to make a recommendation on caffeine restriction and additional evidence from reviews of non-randomized studies (NRSs) was thus evaluated. Two NRS reviews asked the question, “Is there an association between maternal caffeine intake and the risk of low birth weight?” (77, 78), and two reviews asked the question “Is there an association between maternal caffeine intake and the risk of pregnancy loss?” (79, 80). In these reviews, low caffeine intake was defined as less than 150 mg caffeine per day, and high caffeine intake was defined as more than 300 mg or more than 350 mg per day. All four reviews adjusted data for smoking and other variables, and performed dose–response meta-analyses.

#### Fetal and neonatal outcomes: low birth weight

Moderate-certainty evidence from one review shows that high caffeine intake (more than 300 mg) is probably associated with a greater risk of low birth weight than low or no caffeine intake (12 studies; odds ratio [OR]: 1.38, 95% CI: 1.10–1.73) (78). Very low- to moderate-certainty evidence from the other review was stratified according to dose and shows that very low caffeine intake may be associated with fewer low-birth-weight neonates than low (5 studies; RR: 1.13, 95% CI: 1.06–1.21), moderate (7 studies; RR: 1.38, 95% CI: 1.18–1.62) or high caffeine intake (8 studies; RR: 1.60, 95% CI: 1.24–2.08) (77).